



Multi-Material Topology Optimization

with General Convex Polytopal Constraints

Dohyun Kim

joint work with P. Gangl, B. Keith, B. S. Lazarov, T. M. Surowiec

Division of Applied Mathematics
Brown University

MFEM Community Workshop 2026



Collaborators

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Brendan Keith

Applied Mathematics
Brown University



Boyan S. Lazarov

Lawrence Livermore
National Laboratory



Thomas M. Surowiec

Simula Research
Laboratory



Peter Gangl

RICAM
Austrian Academy of Sciences

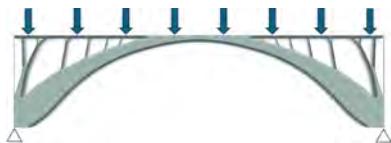
Supported by DOE Office of Science Early Career Research Program (DE-SC0024335) and CIGMO, a PSAAP-IV center funded by DOE/NNSA (DE-NA0004261).



One Material, Then Many

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Where Should Material Go?



$$\min_{\rho} J(\rho) = \int_{\Omega} f \cdot u(\rho) \, dx$$

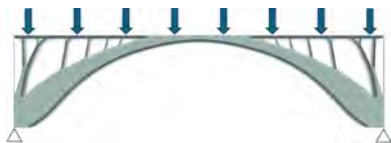
$$\text{s.t. } \rho(x) \in [0, 1], \quad \int_{\Omega} \rho \, dx = V|\Omega|$$

- **Pointwise bound** + one volume budget (linear)
- Gradients: one adjoint solve

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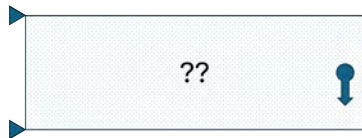


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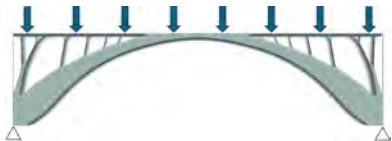


- n phases $\{C_i\}$: stiff / light / cheap, or **fiber angles**
- **One-of- n choice at every point**, per-phase budgets (still linear)
- Relax the discrete choice

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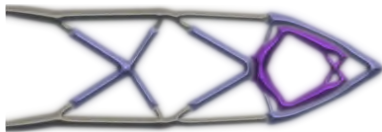


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SiMPL in One Slide: Mirror Descent



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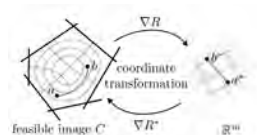
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Projected Gradient $\rightarrow$ Mirror Descent (Two Edits)

given ρ^0 ; for $k = 0, 1, 2, \dots$

state + adjoint solve $\rightarrow g^k = \nabla J(\rho^k)$

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SiMPL in One Slide: Mirror Descent



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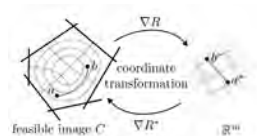
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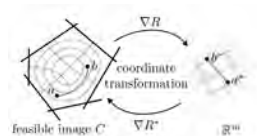
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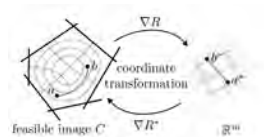
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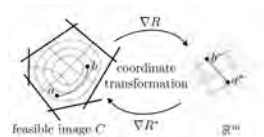
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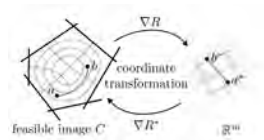
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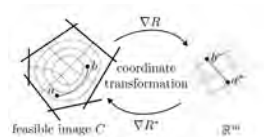
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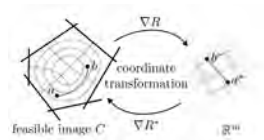
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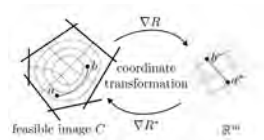
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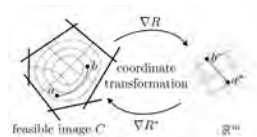
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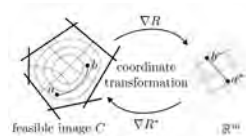
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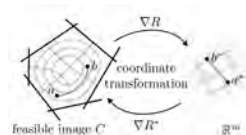
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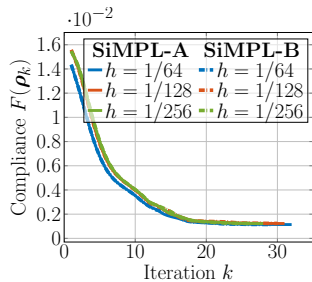
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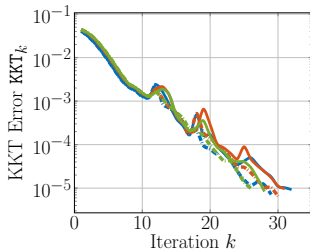
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- Ships as **MFEM Example 37**

What Convergence Looks Like

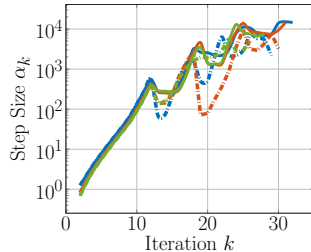
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Compliance $F(\rho^k)$



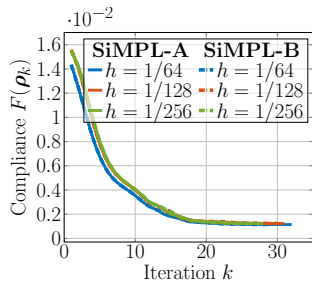
KKT error: tolerance in ~ 30 its



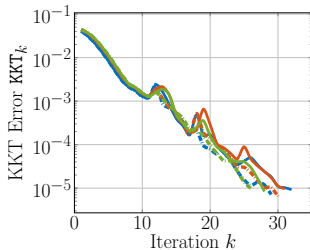
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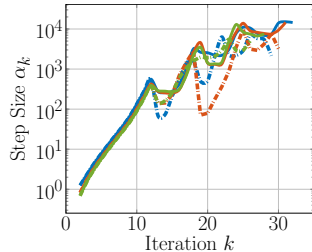
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Curves for $h = 1/64, 1/128, 1/256$ lie **on top of each other**:
mesh-independent iteration counts, measured.

Is the Simplex the Natural Choice?

5 / 18

Conventional: Lift to Fractions

- Weights $\lambda \in \Delta^{n-1}$, design

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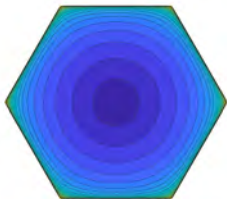


Figure: Materials = vertices of a polytope
 $P = \text{conv}\{v_i\}$

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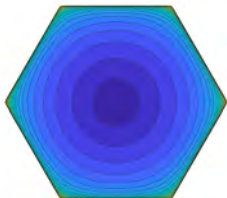


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Orthotropic: The Angles Already Live in 2-D

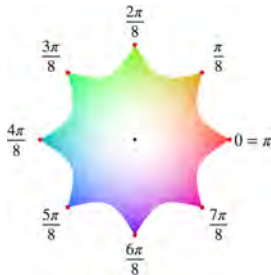


Figure: Double-angle plane: vertex k sits at $2\theta_k$

- More angles = finer polygon,
still dim 2

Entropy Handles Geometry (Hidden in D_R)



Keith & Surowiec

An entropy $R : C \rightarrow \mathbb{R}$ gives a bijection $\nabla R^* : \mathbb{R}^d \rightarrow \text{int } C$.



Box: Fermi–Dirac $\Rightarrow$
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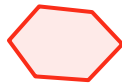
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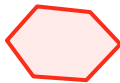
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Key Idea

An entropy for a **general convex polytope** $P = \text{conv}\{v_1, \dots, v_n\} \subset \mathbb{R}^d$, with closed-form ∇R^* .

The Polytopal Entropy

7 / 18

Collect the vertices in $V = [v_1 \cdots v_n]$ and push the Gibbs entropy through V :

$$R^*(\psi) = \log \sum_k \exp(v_k^\top \psi) \implies \rho = \nabla R^*(\psi) = V \operatorname{softmax}(V^\top \psi)$$

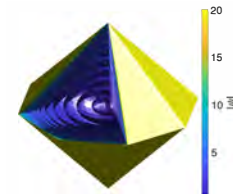


Figure: Octagon-base polytope in $\mathbb{R}^3$: void + orthotropic + isotropic

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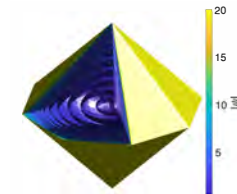


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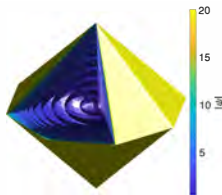


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- Recovers the classics: box $\rightarrow$ sigmoid, simplex $\rightarrow$ softmax

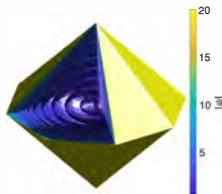


Figure: Octagon-base polytope in $\mathbb{R}^3$: void + orthotropic + isotropic

The Polytopal Entropy

7 / 18

Collect the vertices in $V = [v_1 \cdots v_n]$ and push the Gibbs entropy through V :

$$R^*(\psi) = \log \sum_k \exp(v_k^\top \psi) \implies \rho = \nabla R^*(\psi) = V \operatorname{softmax}(V^\top \psi)$$

- R = Gibbs entropy **inf-projected** onto P
- Closed-form ∇R^* **and** Hessian, cheap at every quadrature point
- Recovers the classics: box $\rightarrow$ sigmoid, simplex $\rightarrow$ softmax

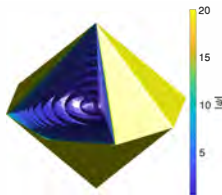


Figure: Octagon-base polytope in $\mathbb{R}^3$: void + orthotropic + isotropic

Intuition

$$|\psi| \rightarrow +\infty \iff \lambda = \operatorname{softmax}(V^\top \psi) \rightarrow e_i \iff \rho \rightarrow \text{vertex}$$

The Algorithm: One Line Changes

8 / 18

SiMPL, Multi-Material Edition

given ψ^0 ; for $k = 0, 1, 2, \dots$

$\rho^k \leftarrow \text{sigmoid}(\psi^k)$

state + adjoint solve $\rightarrow g^k = \nabla J(\rho^k)$

$\alpha_k \leftarrow$ secant curvature guess

$\psi^{k+1} \leftarrow \psi^k - \alpha_k g^k - \mu_k$, backtrack α_k until J decreases

// μ_k : a small dual solve for the global linear constraints

The Algorithm: One Line Changes

8 / 18

SiMPL, Multi-Material Edition

given ψ^0 ; for $k = 0, 1, 2, \dots$

$\rho^k \leftarrow V \text{softmax}(V^\top \psi^k)$ // the only change

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Everything Else Is Inherited

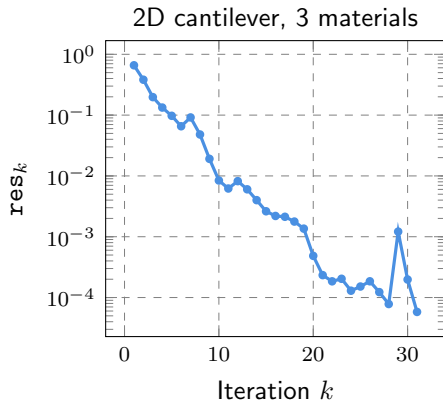
Adaptive step size, monotone backtracking,
KKT gap: all from single-material SiMPL.

- First-order, gradients only
- Feasible at every iterate
- $\psi(x) \in \mathbb{R}^d$, not $\mathbb{R}^n$

When to Stop? Check the Vertices

9 / 18

Step direction $d^k = (\psi^k - \psi^{k+1})/\alpha_k$
 $\approx$ reduced gradient).



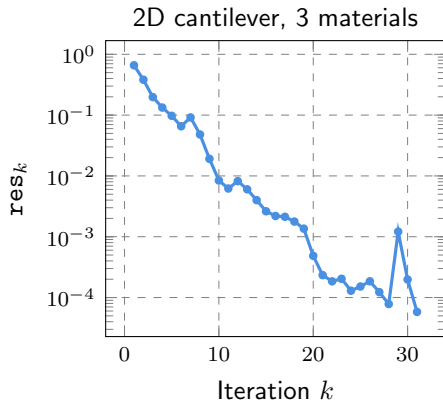
When to Stop? Check the Vertices

9 / 18

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- At each x : could a **pure material** do better?

$$\text{res}_k = \int_{\Omega} \left| \min_{v \in P} d^k \cdot (v - \rho^k) \right| dx$$



When to Stop? Check the Vertices

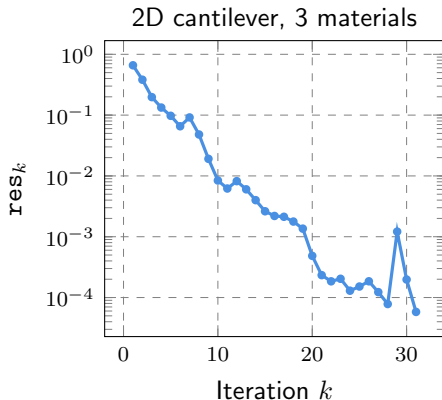
9 / 18

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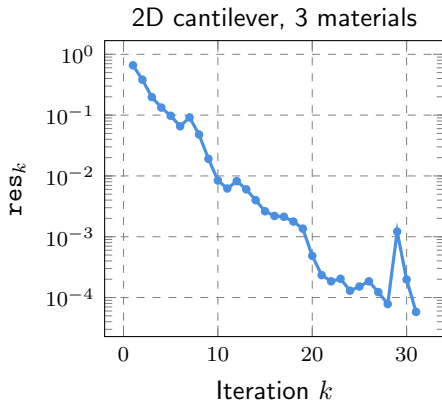
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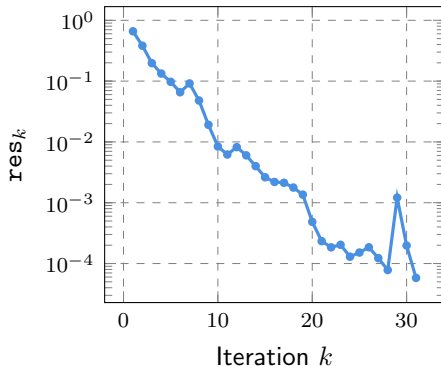
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In MFEM

A pointwise Coefficient (q dot products),
ProjectCoefficient + ComputeL1Error

2D cantilever, 3 materials

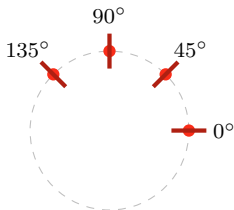


Fiber Angles Live on a Half Circle

10 / 18

Stiffness does not see the direction of a fiber: $C(\theta) = C(\theta + 180^\circ)$.

Naive: Vertex at Angle θ



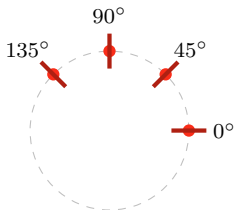
- Vertices crowd one half of the circle
- 0° and 135° look far apart, though they are 45° apart physically

Fiber Angles Live on a Half Circle

10 / 18

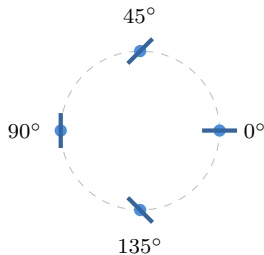
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- 0° and 135° look **far apart**, though they are 45° apart physically

Half-Angle: Vertex at Angle 2θ



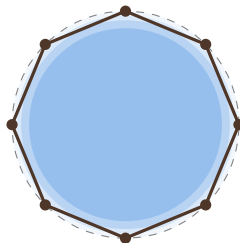
- **Regular polygon**, neighbors stay neighbors
- Read the fiber back as $\theta = \phi/2$
- More angles $\Rightarrow$ finer polygon, still **2 unknowns**

Why the Design Picks a Pure Angle

11 / 18

Write the in-plane design (a, b) in polar form (r, ϕ) :

$$C(\eta) = s^p C_{\text{iso}} + r^p C_{\text{ani}}\left(\frac{\phi}{2}\right), \quad p = 4$$



dashed: $r = 1$

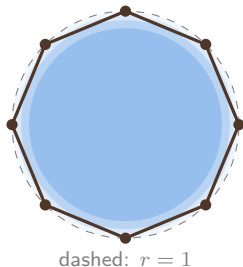
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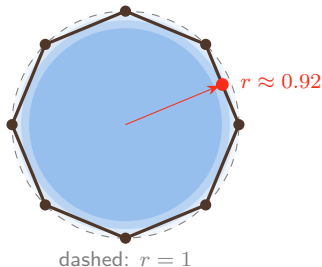
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- 50/50 blend of two neighbors (8 angles):
 $r = \cos 22.5^\circ \approx 0.92$, $r^4 \approx 0.73$
 $\Rightarrow$ same cost, **27% less stiffness**



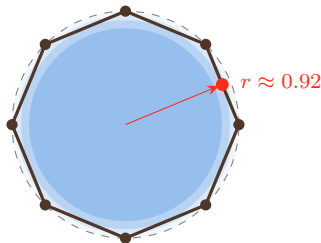
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11 / 18

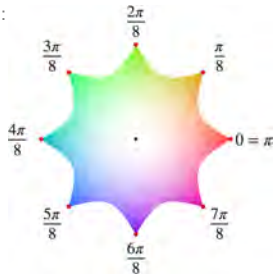
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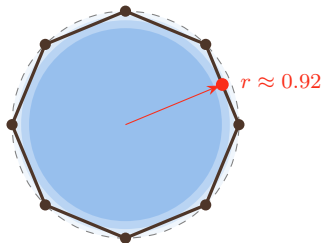


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11 / 18

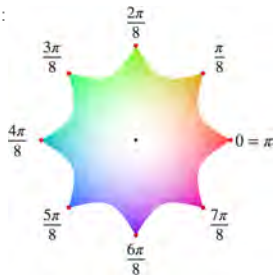
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- The material budget then pushes η to a vertex
- $p = 4$ is chosen so that $|dC/d\eta| < \infty$ at the origin

dashed:



In MFEM: From ex37 to Multi-Material



MFEM ex37

```
constexpr int N = 6; // number of materials
constexpr int MAT_DIM = 2; // dimension of the material space
const future::tensor<MAT_DIM, N> V = ...;
// latent field: Vector-valued QuadratureFunction

for (int k = 0; ; k++) {
    // point kernel: rho = V softmax(V^T psi)
    polytope_map(psi, V, rho);

    // Forward and adjoint solves
    forward_solver.Mult(rho, frho_u); // FE state solve
    adjoint_solver.Mult(rho, grad);   // dJ/drho -> L2

    psi.Add(-alpha, grad);           // mirror step
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- ψ in L^2 or QF
⇒ **local, DOF-wise** update
- ∇R^* is a **point kernel**
- $\mu \in \mathbb{R}^m$ solves
a smooth convex dual problem
- Essentially the same cost with
gradient descent

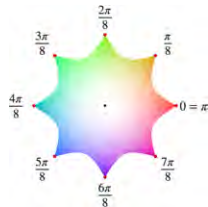
In MFEM: Autodiff for Orthotropy

13 / 18

Design $\eta = (s, a, b)$: solid fraction s + a point (a, b) in the double-angle polygon.

```
// C(eta): rotated + penalized stiffness, written once
class SIMPOrthotropicMaterial : public ADVectorFunction {
  AD_VEC_IMPL(T, VEC, MAT, eta, C_rot, {
    const T s = eta[0], a = eta[1], b = eta[2];
    // ... rotation algebra in T
  });
};

// C(eta) enters as a MatrixCoefficient
a.AddDomainIntegrator(
  new OrthotropicElasticityIntegrator(C_cf));
```

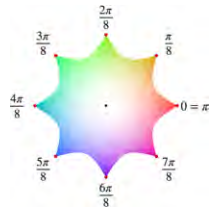


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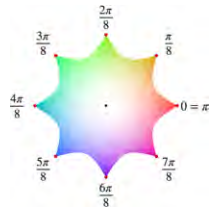
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In MFEM: Autodiff for Orthotropy

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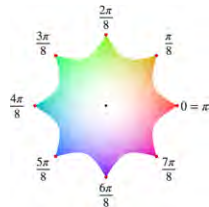
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- Gradient **without hand-derived sensitivities**

In MFEM: Autodiff for Orthotropy

13 / 18

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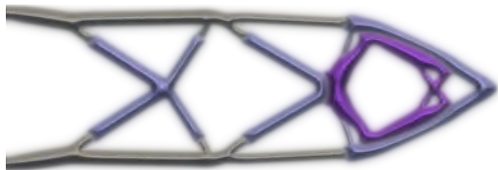
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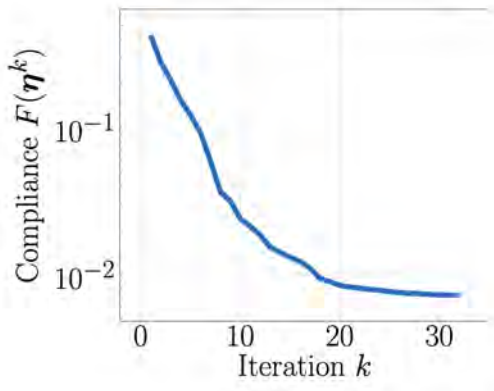
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- Gradient **without hand-derived sensitivities**
- Migration to ∂ FEM

Results: 2D Multi-Material

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Three materials, per-phase budgets
(**31 iterations**)



Compliance history: **monotone descent**

Results: 3D and Orthotropic Compliance

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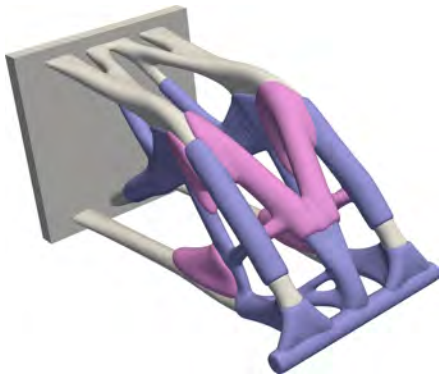


Figure: 3D, three materials (**47 iterations**)

Results: 3D and Orthotropic Compliance

15 / 18

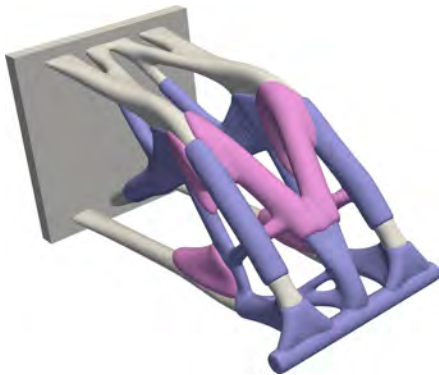


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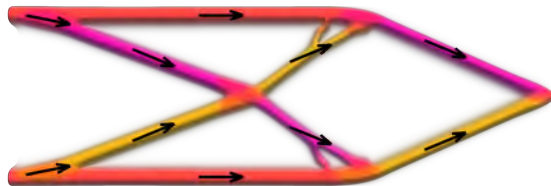


Figure: Orthotropic cantilever, 8 discrete fiber angles; color = angle (**35 iterations**)

Continuous Angles: The Limit of the Polygon

16 / 18

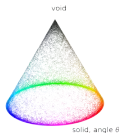


Figure: Cone = latent ball mapped by ∇R^* : rim = solid at **any** θ , apex = void



Figure: Converged design with **continuous** fiber angle (color), volume fraction 0.5, $h = 1/256$

Continuous Angles: The Limit of the Polygon

16 / 18

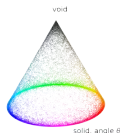


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Mesh size h	Control unknowns	Iterations	Compliance	KKT measure
1/64	36,864	28	2.76×10^{-3}	9.1×10^{-5}
1/128	147,456	30	2.77×10^{-3}	9.9×10^{-5}
1/256	589,824	31	2.77×10^{-3}	9.7×10^{-5}

Permanent Magnet Placement

17 / 18

Maximize magnetic flux through the airgap:

$$\max \int_{\Gamma} \text{curl } u \cdot \mathbf{n} \, ds$$

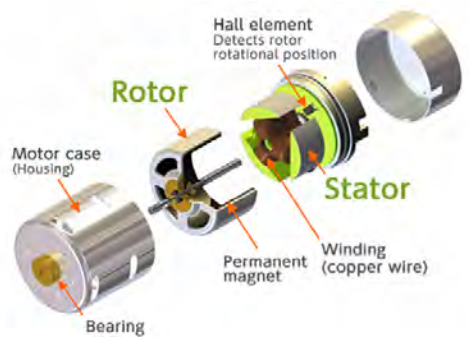


Figure: The device; the rotor is the design domain

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17 / 18

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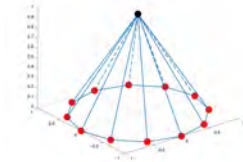


Figure: Red = permanent magnet direction, Black = iron

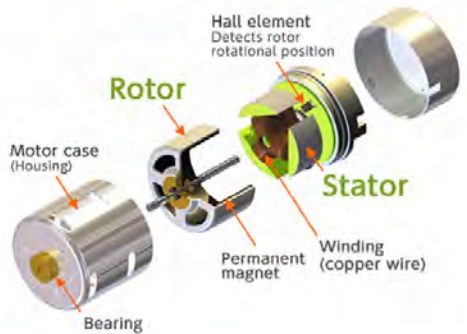


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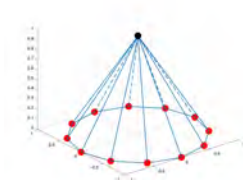


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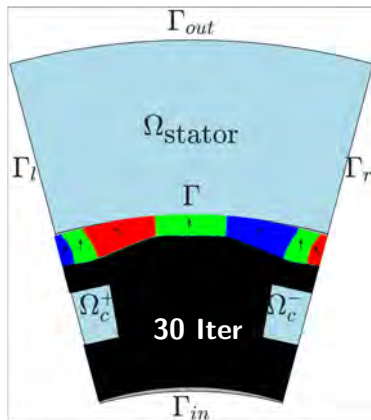


Figure: Optimized rotor: material + orientation per point

Conclusion

18 / 18

SiMPL for physics-encoded polytopal constraints
Same algorithm, different pointwise constraint set.



Prox. Galerkin



MFEM ex37



SIOPT '25



SMO '25



this work

Conclusion

18 / 18

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- The loop never changes; only ∇R^* is swapped



Prox. Galerkin



MFEM ex37



SIOPT '25



SMO '25



this work

Conclusion

18 / 18

SiMPL for physics-encoded polytopal constraints

Same algorithm, different pointwise constraint set.

- The loop never changes; only ∇R^* is swapped
- Box, simplex, polytope, orientation polygon: **one framework**, feasible by construction



Prox. Galerkin



MFEM ex37



SIOPT '25



SMO '25



this work

Conclusion

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SiMPL for physics-encoded polytopal constraints

Same algorithm, different pointwise constraint set.

- The loop never changes; only ∇R^* is swapped
- Box, simplex, polytope, orientation polygon: **one framework**, feasible by construction
- **Mesh-independent** iterations, first-order cost



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General global constraints (e.g. min. volume s.t. critical compliance)

Robust dual solver

Black-box optimizer w/ pointwise convex constraints



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